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Session : Name of the Session

Reimagining Distribution Grid Planning: Integrating Prosumers, EVs, and DERs through Integrated Distribution Resource Planning

Presented By – Dr. Adarsh Nagarajan

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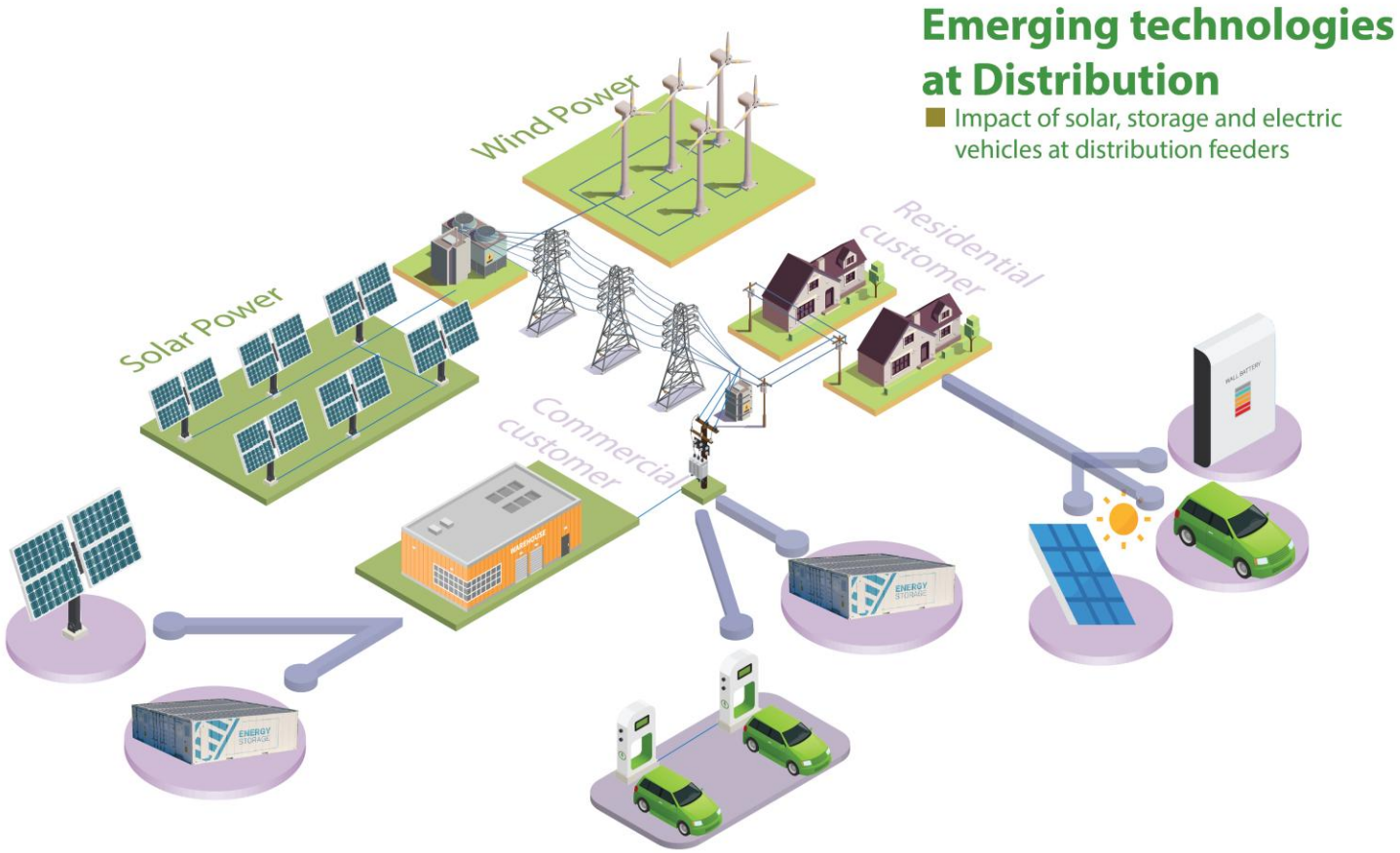
INTRODUCTION

- Indian DISCOMs are evolving from passive to bidirectional platforms.
- Fueled by aggressive renewable energy targets and electrification mandates.
- Traditional capital expenditure (capex) planning, rooted in deterministic peak forecasting and "wires-only" reinforcements, is misaligned with this present needs.



Background

- DISCOMs allocated ₹78,000 crore in capex, with 58% (₹45,000 crore) to augmentation, amid ₹7.4 lakh crore debt and 15.8% AT&C losses equating to ₹0.21/kWh excess costs

- This paper posits that Integrated **Distribution Resource Planning (DRP)**—a **granular, scenario-based methodology**—integrates **prosumers, EVs, and DERs as grid assets**, valuing their locational and temporal benefits via cost-benefit analysis

Traditional spending plan approach

- At present CAPEX focuses on conventional network upgrades – transformers, conductors
- In FY23-24, augmentation/reinforcement dominated at 58% (₹45,000 crore), including lines, transformers, and substations [7].
- Sizing conventional reinforcements without accounting for DER impact risks misallocating capital.
- DSM/flexibility received <4% (₹3,000 crore), underscoring underinvestment in DER integration [14].

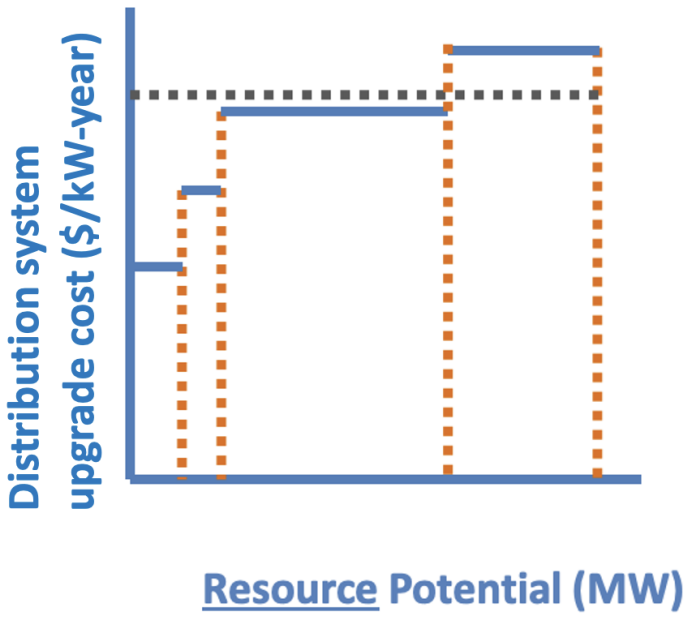
Category	Share (%)	Amount (₹ Cr)	Key Focus
Augmentation/ Reinforcement	58	45,000	Lines, transformers
Metering/IT	18	14,000	Smart meters, SCADA
DSM/Flexibility	4	3,000	Demand response, storage
Others (Admin, Misc.)	20	16,000	Consumer assets, R&M

`FY	Total Capex	Augmentation Share (%)
2014	20,000	40
2019	45,000	50
2024	78,000	58

Limitations of existing CAPEX approaches

- Non-linear Network Costs
- Rising Software and Technology Needs
- Power Purchase Avoidance Not Credited
- Demand Flexibility Not Considered

Overestimation - Traditional models rely on rating-based upgrades, ignoring hosting capacity impacts and DER variability.



EE + Electrification (Permanent DSM)	Passive Demand Response (e.g., TOU rates)	Active Demand Response (e.g., DR program)	Energy Storage	Distributed Generation
• Heat Pumps – Residential • Heat Pumps - C&I • District Heating • Electric Vehicles (EVs) • Water Heating/HPWH • Envelope/ Weatherization • Conservation Voltage Reduction	• Smart Thermostats • EV Managed Charging • C&I Demand Response	• Smart Thermostats • EV Managed Charging • C&I Demand Response	• Stationary Batteries - BTM • Stationary Batteries - FOM • EV batteries - V2X • Thermal storage	• Solar PV - BTM • Solar PV - Community • Combined Heat & Power (CHP) • NG/Diesel Generators • Distributed Wind

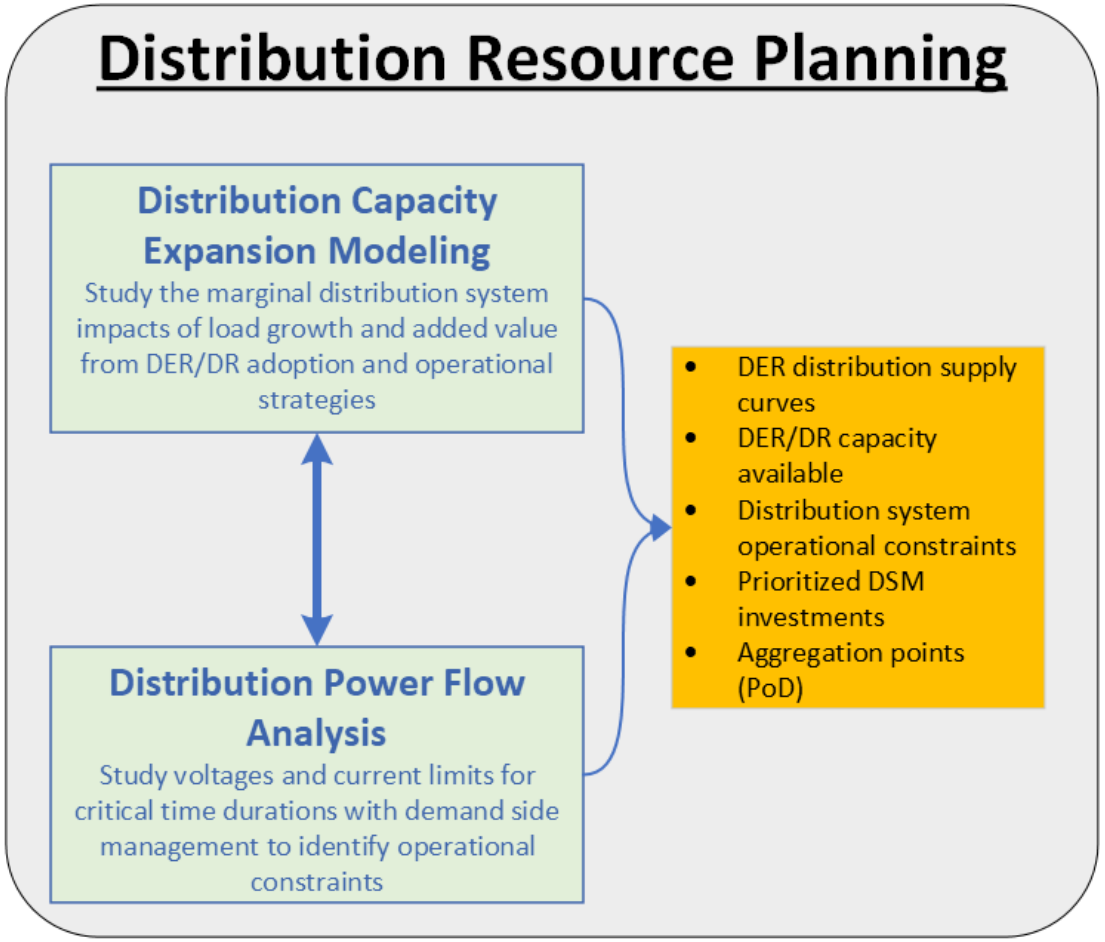
Distribution Resource Planning (DRP) – A solution

Key outcomes from DRP include:

- Identification of feeder and substation constraints
- Hosting capacity limits for DER
- Available flexibility from demand response or storage
- Comparison of network upgrades versus DER alternatives

The economic benefit can be expressed as:

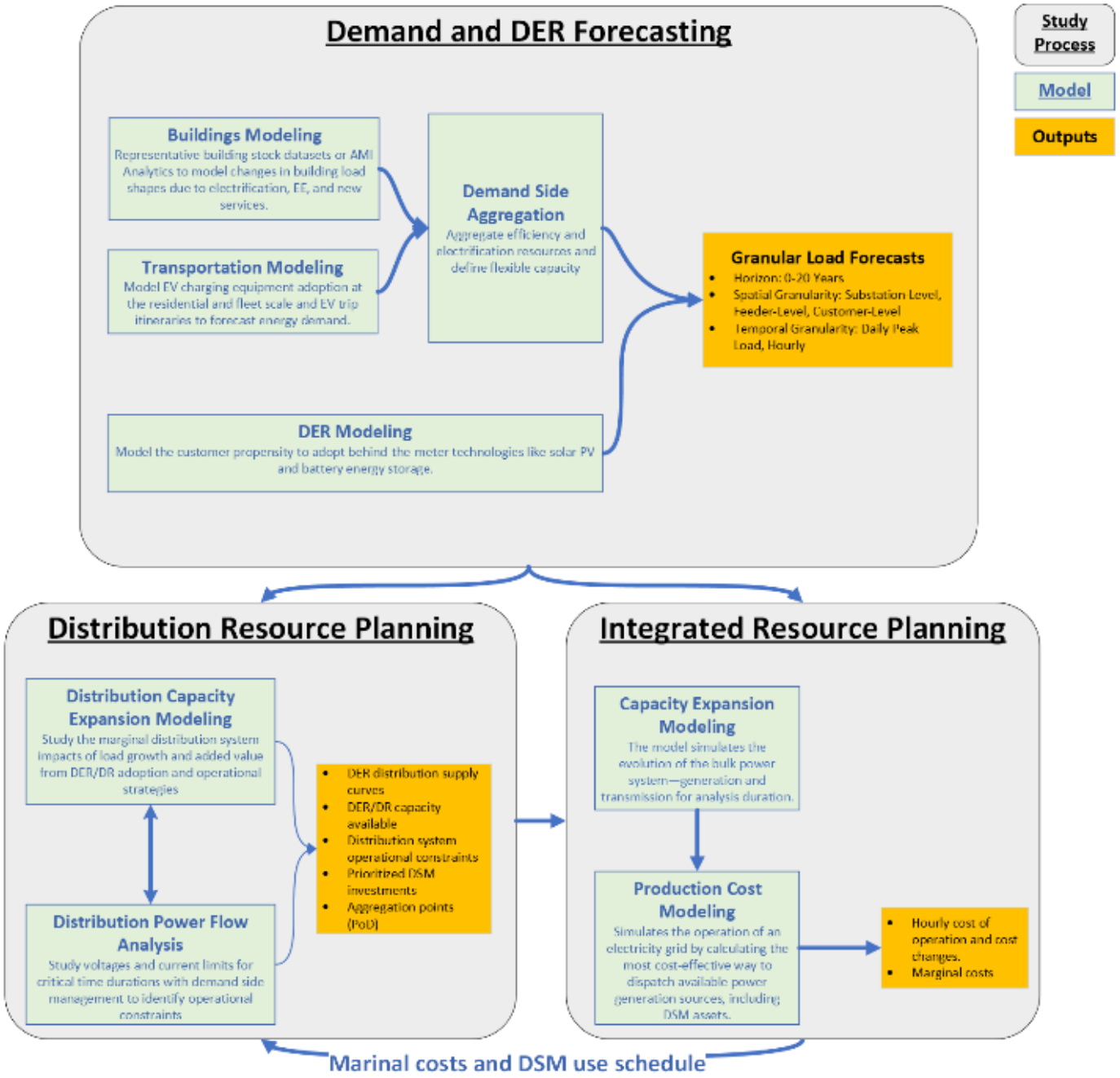
$$\text{Benefit} = \int (\text{Avoided Network Cost} - \text{DER Cost})\, dt$$



DRP helps utilities **plan smarter investments, reduce peak stress on the grid, and integrate distributed energy resources while maintaining reliability**.

Integrated Systems Planning

Approach	Description	Primary Benefit
Distribution Priority – 11kV and below	DRP optimizes local constraints; IRP receives net load after DER actions	Maximizes local hosting capacity and capex deferral
Bulk System Priority – 33kV and above	IRP optimizes supply stack; DRP adapts local operations	Captures system-wide cost of flexibility
Coordinated Optimization	Unified process balancing distribution and bulk needs	Most efficient totex outcome



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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

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Links/References (If any)